MILITARY SPECIFICATION

ISOPROTERENOL HYDROCHLORIDE AND PHENYLEPHRINE BITARTRATE AEROSOL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Isoproterenol Hydrochloride and Phenylephrine Bitartrate Aerosol (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a nonethanolic, nonaqueous, pressurized aerosol containing a suspension of micronized Isoproterenol Hydrochloride, 4 mg, and Phenylephrine Bitartrate, 6 mg, per ml. The inert propellent shall be a mixture of fluorochlorohydrocarbons. The aerosol may contain a suitable surfactant as a dispersing agent.

3.1.1 Assay. The aerosol shall contain in each milliliter 4.0 mg ± 10 percent Isoproterenol Hydrochloride and 6.0 mg ± 10 percent Phenylephrine Bitartrate when determined as specified in 4.7.1.

3.1.2 Adapter delivery. Each dose delivered through the oral adapter shall contain 0.16 mg ± 20 percent Isoproterenol Hydrochloride and 0.24 mg ± 20 percent Phenylephrine Bitartrate when determined as specified in 4.7.2.

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3.1.3 Minimum number of doses. The aerosol shall deliver 450 measured doses through the oral adapter when determined as specified in 4.7.3.

3.1.4 Particle size. The great majority of individual particles shall be less than 5 µm in diameter, and no more than 10 individual crystalline particles shall be more than 10 µm in the longest dimension, when determined under "Particle Size" in the NF monograph for "Isoproterenol Sulfate Aerosol".

3.1.5 Moisture. The inhalation shall have a total moisture content of not more than 150 parts per million (p.p.m.) when determined as specified in 4.7.4.

3.2 Ingredients.

3.2.1 Isoproterenol hydrochloride. The isoproterenol hydrochloride used in the manufacture of the aerosol shall be in accordance with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.2.2 Phenylephrine bitartrate. The phenylephrine bitartrate used in the manufacture of the suspension shall comply with the following:

3.2.2.1 Identification. Shall comply with the identification tests specified in 4.7.5.1.

3.2.2.2 Residue on Ignition. Shall be no more than 0.2 percent when determined by the USP method.

3.2.2.3 Loss of Drying. Shall be not more than 0.5 percent when dried in vacuum over silica gel for 4 hours.

3.2.2.4 Assay. Shall assay not less than 97.0 percent and not more than 103.0 percent on a dried basis when determined as specified in 4.7.5.2.

3.2.3 Other ingredients. All other ingredients entering into the finished aerosol shall be of USP or NF quality or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

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3.3 Expiration dating period. The expiration dating period shall be 60 months.

3.3.1 Delivery. Not more than 6 months of the expiration dating period shall have elapsed at time of delivery to the Government.

3.4 Workmanship. Workmanship shall be first class throughout. The component parts of the completed inhaler assembly shall be free from flaws or other defects which detract from their appearance or which may impair the function of the assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of aerosol from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

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[4] Label adhesion test shall not apply.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Unit of issue
Minor	II	2.5	Unit of issue

4.3.1 Raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay	3.1.1	S-2	-
Adapter delivery	3.1.2	S-2	-
Minimum number of doses	3.1.3		
Particle size	3.1.4	S-2	-
Moisture	3.1.5	S-2	-
Ingredients: 2/			
Isoproterenol Hydrochloride	3.2.1		
Phenylephrine Bitartrate	3.2.2		

1/ Where possible, the sample shall be used for two or more tests.
2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. The finished aerosol shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

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4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Valve does not function properly (continuous or no spray).
102	Valve stem is loose in adapter.
103	Adaptor is clogged (does not spray).
104	Adaptor not free of cracks or breaks.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.
203	Adaptor not clean.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Assay. Part I, Isoproterenol Hydrochloride Content (Thru Valve).

Reagents

USP Isoproterenol Hydrochloride Reference Standard.
Sodium Bisulfite solution, 0.2 percent (w/v) - Dissolve 1 gram of sodium bisulfite in 500 ml of purified water, and mix.
Chloroform, reagent grade.
Ferrous sulfate reagent - To a 200 ml volumetric flask add 1.0 ml of 1.0N hydrochloric acid, 100 ml of purified water, 1,000 mg of sodium bisulfite and 1,500 mg of ferrous sulfate. Dissolve and bring to volume with purified water, and mix.

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Buffer - To about 720 ml of purified water in a 2,000 ml volumetric flask, add 168 grams of sodium bicarbonate and 200 grams of potassium bicarbonate (not all solids will dissolve at this stage). To another 720 ml of purified water add 150 grams of aminoacetic acid and 68 ml of ammonium hydroxide, and mix to dissolve. Add this solution to the first, bring to 2,000 ml with purified water, and mix until solution is complete.

Mixed reagent - Add in this order, 500 mg of sodium citrate to 10.0 ml of ferrous sulfate reagent, then add 100 ml of buffer, and mix.

Procedure.

1. Weigh accurately approximately 60 mg of USP Isoproterenol Hydrochloride Reference Standard into a 100 ml volumetric flask. Dissolve and bring to volume with 0.2 percent sodium bisulfite solution, and mix. Pipet 10.00 ml of this solution into a 100 ml volumetric flask, dilute to volume with 0.2 percent sodium bisulfite solution, and mix. This is the "standard".
2. Weigh accurately each of 8 aerosols, and record (fill weight).
3. Place 20.00 ml of chloroform in each of eight 100 ml beakers.
4. Prime the aerosols by alternately shaking and firing them 10 times through an appropriate acutator, and wipe clean the valve stem and ferrule.
5. Reweigh accurately the aerosols, and record.
6. Treat each primed aerosol identically as follows:
   a. Hold the aerosol, valve stem down and slightly above the chloroform in a 100 ml beaker.
   b. Shake gently, and carefully lower the aerosol along side of the beaker until the valve stem touches the bottom. Press down firmly to deliver a single spray, then slowly release the pressure.
   c. Repeat steps a. and b. for a total of 3 sprays, and then shake gently to empty the valve stem and transfer any residue to the beaker.
   d. Rinse the valve stem and ferrule with about 2 ml of chloroform, collecting the rinse in the beaker.
   e. Allow the aerosol to dry, reweigh accurately, and record.
7. Treat each chloroform solution identically as follows:
   a. Swirl carefully, and transfer quantitatively to a 50 ml centrifuge tube.
   b. Rinse the beaker with two 3 ml portions of chloroform, and add the rinses to the centrifuge tube.
   c. Add 10.00 ml of 0.2 percent sodium bisulfite to the centrifuge tube, stopper, and shake vigorously for 1 minute.
   d. Centrifuge the mixture for 20 minutes. The clear supernate is a "sample".
8. Treat the "samples" and "standard" (duplicate) identically as follows:
   a. Pipet 5.00 ml into a 25 ml Erlenmeyer flask.
   b. Add 1.10 ml of mixed reagent, and mix.
9. Measure and record the absorbances at 530 nm of the "treated samples" and "treated standard" (duplicate) on a suitable spectrophotometer using matched, 1.00 cm cells with purified water as referenced solvent.
10. Calculate the isoproterenol hydrochloride content for each "sample" from the following formula, record and report.

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Isoproterenol hydrochloride content (mg/ml in aerosol) =

$$\frac{(Asam)\ (Wstd)\ (1.40)\ \ (10)}{(Astd)\ (1000)\ \ \ \ \ \ (SW)}$$

Asam, Astd = absorbances at 530 nm of the "treated sample" and the "treated standard" (average of duplicates), respectively.
Wstd = weight of isoproterenol hydrochloride taken for "standard" (mg).
SW = total weight of aerosol formulation delivered in the 3 sprays (weight in step 5 less weight in step 6e) (gram).
1.40 = density of aerosol formulation (gram/ml).
10 = total volume of the "sample" (ml).
1000 = equivalent total volume of the "standard" (ml).

Part II, Phenylephrine Bitartrate Content (Thru Valve)

Reagents

USP Phenylephrine Hydrochloride Reference Standard.
Sulfuric acid, 0.05N - Dilute 1.4 ml of sulfuric acid to 1000 ml with purified water, and mix.
Sulfuric acid, 5N - Dilute 140 ml of sulfuric acid to 1000 ml with chilled purified water, and mix.
Chloroform, reagent grade.
Mercuric sulfate reagent - Dissolve 150 gram of yellow mercuric oxide in 800 ml of purified water, and while stirring, slowly add 200 ml of sulfuric acid.
Sodium nitrite reagent, 1.2 percent (w/v) - Prepare fresh daily. Dissolve 120 mg of sodium nitrite in 10 ml of purified water, and mix.

Procedure

1. Weigh accurately approximately 60 mg of USP Phenylephrine Hydrochloride Reference Standard into a 100 ml volumetric flask. Dissolve and bring to volume with 0.05N sulfuric acid, and mix. Pipet 10.00 ml of this solution into a 50 ml volumetric flask, dilute to volume with 0.05N sulfuric acid, and mix. This is the "standard".
2. Weigh accurately each of 8 aerosols, and record (fill weight).
3. Place 20.00 ml of chloroform in each of eight 100 ml beakers.
4. Prime the aerosols by alternately shaking and firing them 10 times through an appropriate actuator.
5. Reweigh accurately the aerosols, and record.

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6. Treat each primed aerosol identically as follows:
   a. Hold the aerosol, valve stem down and slightly above the chloroform in a 100 ml beaker.
   b. Shake gently, and carefully lower the aerosol along side of the beaker until the valve stem touches the bottom. Press down firmly to deliver a single spray, then slowly release the pressure.
   c. Repeat steps a. and b. for a total of 6 sprays, and then shake gently to empty the valve stem and transfer any residue to the beaker.
   d. Rinse the valve stem and ferrule with about 2 ml of chloroform, collecting the rinse in the beaker.
   e. Allow the aerosol to dry, reweigh accurately, and record.
7. Treat each chloroform solution identically as follows:
   a. Swirl carefully, and transfer quantitatively to a 50 ml centrifuge tube.
   b. Rinse the beaker with two 3 ml portions of chloroform, and add the rinses to the centrifuge tube.
   c. Add 10.00 ml of 0.05N sulfuric acid to the centrifuge tube, stopper, and shake vigorously for 1 minute.
   d. Centrifuge the mixture for 20 minutes. The clear supernate is a "sample".
8. Treat the "samples", "standard" (duplicate) and 0.05N sulfuric acid "blank" identically as follows:
   a. Pipet 5.00 ml into a 50 ml centrifuge tube.
   b. Add 3.00 ml of mercuric sulfate reagent, and mix.
   c. Place in a water bath at 37°C for 10 minutes.
   d. Allow to stand at room temperature for 30 minutes.
   e. Add 250 µl of sodium nitrite reagent, and mix by swirling occassionally for 30 minutes.

   These are the "treated" "samples", "standard" and "blank".
9. Measure and record the absorbances at 495 nm of the "treated samples" and "treated standard" (duplicate) on a suitable spectrophotometer using matched, 1.00 cm cells with the "treated blank" as referenced solvent.
10. Calculate the phenylephrine bitartrate content for each "sample" from the following formula, record and report.

Phenylephrine bitartrate content (mg/ml in aerosol) =

$$\frac{(Asam)\ (Wstd)\ (1.56)\ (1.40)\ (10)}{(Astd)\qquad (SW)\qquad (500)}$$

Asam, Astd = absorbances at 495 nm of the "treated sample" and the "treated standard" (average of duplicates), respectively.
Wstd = weight of phenylephrine hydrochloride taken for the "standard" (mg).
SW = total weight of aerosol formulation delivered in the 6 sprays (weight in step 5 less weight in step 6e) (gram).
1.40 = density of aerosol formulation (g/ml).
10 = total volume of the "sample" (ml).
500 = equivalent total volume of the "standard" (ml).
1.56 = gravimetric factor, phenylephrine hydrochloride to phenylephrine bitartrate.

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4.7.2 Adapter Delivery. Medication Delivery (Thru Adapter), Part I, Isoproterenol Hydrochloride Delivered.

Specialized Apparatus

Shall be as described under "Unit Spray Sampling Apparatus" (Artificial Respiratory System (ARS)) in the NF section entitled "Aerosols". Alternately, an apparatus embodying the principles of the designated apparatus may be used.

Reagents

USP Isoproterenol Hydrochloride Reference Standard.
Sodium bisulfite solution, 0.2 percent (w/v) - Dissolve 1 gram of sodium bisulfite in 500 ml of purified water, and mix.
Chloroform, reagent grade.
Ferrous sulfate reagent - To a 200 ml volumetric flask add 1.0 ml of 1.0N hydrochloric acid, 100 ml of purified water, 1000 mg of sodium bisulfite and 1,500 mg of ferrous sulfate. Dissolve and bring to volume with purified water, and mix.
Buffer - To about 720 ml of purified water in a 2,000 ml volumetric flask, add 168 gram of sodium bicarbonate and 200 gram of potassium bicarbonate. To another 720 ml of purified water add 150 gram of aminoacetic acid and 68 ml of ammonium hydroxide, and mix to dissolve. Add this solution to the first, bring to 2,000 ml with purified water, and mix until solution is complete.
Mixed reagent - Add in this order, 500 mg of sodium citrate to 10.0 ml of ferrous sulfate reagent, then add 100 ml of buffer, and mix.
95 percent Ethanol (Alcohol, USP).

Procedure

1. Weigh accurately approximately 60 mg of USP Isoproterenol Hydrochloride Reference Standard into a 100 ml volumetric flask. Dissolve and bring to volume with 0.2 percent sodium bisulfite solution, and mix. Pipet 10.00 ml of this solution into a 100 ml volumetric flask, dilute to volume with 0.2 percent sodium bisulfite solution, and mix. This is the "standard".
2. Treat each of 2 aerosols identically as follows:
   a. Pipet 20.00 ml of 0.2 percent sodium bisulfite solution into the gas-washing bottle cylinder of the ARS, and seat it in position.
   b. Prime the aerosol by alternately shaking and firing it 10 times through an appropriate actuator.
   c. Couple the aerosol vial and actuator to the intake tube of the ARS, turn on the vacuum source, and set the flow regulator for an air flow rate of 12 ± 1 liters per minute. Turn off the vacuum, and remove the vial, leaving the actuator in place.
   d. Weigh accurately the aerosol vial, and record.
   e. Deliver a total of 6 sprays into the ARS by repeatedly shaking the vial, seating it in the actuator, turning on the vacuum, immediately delivering a single spray, turning off the vacuum, and separating the vial from the actuator (count sprays accurately).
   f. Uncouple the aerosol vial and actuator from the intake tube of the ARS, and clamp the intake tube in a vertical position.

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g. Weigh accurately the aerosol vial, and record. Substract this weight from that found in step 2d. to verify the number of sprays delivered.

3. Scavenge the drug substance from the ARS in the following manner:
   a. Remove the gas-washing bottle cylinder, and transfer the sample solution to a small beaker.
   b. Pour the sample solution down the wall of the inlet tube, collecting it in the gas-washing bottle cylinder placed a few cm below its normal position in the ARS.
   c. Recycle the sample solution twice more by the application of steps 3a. and 3b.
   d. Transfer the sample solution to a 50 ml centrifuge tube, add 10 ml of chloroform, stopper, and shake vigorously for 1 minute.
   e. Centrifuge the mixture for 20 minutes. The clear supernate is a "sample".
4. Prepare the ARS for subsequent sampling by the following procedure:
   a. Using a washbottle, rinse the wall of the intake tube with about 25 ml of 95 percent ethanol, collecting the rinse in the gas-washing bottle cylinder placed a few cm below its normal position.
   b. Seat the cylinder in its normal position, and turn on the vacuum briefly, drawing any residual rinse through the bubbler and into the cylinder.
   c. Remove the cylinder, and discard the rinse.
   d. Repeat steps 4a. through 4c. with about 25 ml of purified water.
   e. Repeat steps 4a. through 4c. with about 25 ml of 95 percent ethanol.
   f. Seat the cylinder in its normal position, and turn on the vacuum to "dry" the inlet tube, bubbler tube and bubbler.
   g. Replace the cylinder with a clean, dry one, and return the intake tube to its normal horizontal position. The ARS is now ready for the next sample.
5. Treat the "samples" and "standard" (duplicate) identically as follows:
   a. Pipet 5.00 ml into a 25 ml Erlenmeyer flask.
   b. Add 1.10 ml of mixed reagent, and mix.

   These are the "treated" "samples" and "standard".
6. Measure and record the absorbances at 530 nm of the "treated samples" and "treated standard" (duplicate) on a suitable spectrophotometer using matched, 1.00 cm cells with purified water as reference solvent.
7. Calculate the isoproterenol hydrochloride delivered for each "sample" from the following formula, record and report.

Isoproterenol hydrochloride (mg/spray) =

$$\frac{(Asam)\ (Wstd)\ (20)}{(Astd)\ (1000)\ (6)}$$

Asam, Astd = absorbances at 530 nm of the "treated sample" and the "treated standard" (average of duplicates), respectively.
Wstd = weight of isoproterenol hydrochloride taken for the "standard" (mg).
20 = total volume of the "sample" (ml).
1000 = equivalent total volume of the "standard" (ml).
6 = number of sprays taken for the "sample" (spray).

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Part II, Phenylephrine Bitartrate Delivered.

Reagents

USP Phenylephrine Hydrochloride Reference Standard.
Sulfuric acid, 0.05N - Dilute 1.4 ml of sulfuric acid to 1,000 ml with purified water, and mix.
Sulfuric acid, 5N - Dilute 140 ml of sulfuric acid to 1,000 ml with chilled purified water, and mix.
Chloroform, reagent grade.
Mercuric sulfate reagent - Dissolve 150 gram of yellow mercuric oxide in 800 ml of purified water, and while stirring, slowly add 300 ml of sulfuric acid.
Sodium nitrite reagent - 1.2 percent (w/v) - Prepare fresh daily. Dissolve 120 mg of sodium nitrite in 10 ml of purified water, and mix.
95 percent Ethanol (Alcohol, USP).

Procedure

1. Weigh accurately approximately 60 mg of USP Phenylephrine Hydrochloride Reference Standard into a 100 ml volumetric flask. Dissolve and bring to volume with 0.05N sulfuric acid, and mix. Pipet 10.00 ml of this solution into a 50 ml volumetric flask, dilute to volume with 0.05N sulfuric acid, and mix. This is the "standard".
2. Treat each of 2 aerosols identically as follows:
   a. Pipet 20.00 ml of 0.05N sulfuric acid solution into the gas-washing bottle cylinder of the ARS, and seat it in position.
   b. Prime the aerosol by alternately shaking and firing it 10 times through an appropriate actuator.
   c. Couple the aerosol vial and actuator to the intake tube of the ARS, turn on the vacuum source, and set the flow regulator for an air flow rate of 12 $\pm$ 1 liters per minute. Turn off the vacuum, and remove the vial, leaving the actuator in place.
   d. Weigh accurately the aerosol vial, and record.
   e. Deliver a total of 12 sprays into the ARS by repeatedly shaking the vial, seating it in the actuator, turning on the vacuum, immediately delivering a single spray, turning off the vacuum, and separating the vial from the actuator (count sprays accurately).
   f. Uncouple the aerosol vial and actuator from the intake tube of the ARS, and clamp the intake tube in a vertical position.
   g. Weigh accurately the aerosol vial, and record. Substract this weight from that found in step 2d. to verify the number of sprays delivered.
3. Scavenge the drug substance from the ARS in the following manner:
   a. Remove the gas-washing bottle cylinder, and transfer the sample solution to a small beaker.
   b. Pour the sample solution down the wall of the inlet tube, collecting it in the gas-washing bottle cylinder placed a few cm below its normal position in the ARS.

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c. Recycle the sample solution twice more by the application of steps 3a. and 3b.
d. Transfer the sample solution to a 50 ml centrifuge tube, add 10 ml of chloroform, stopper, and shake vigorously for 1 minute.
e. Centrifuge the mixture for 20 minutes. The clear supernate is a "sample".

4. Prepare the ARS for subsequent sampling by the following procedure:
a. Using a washbottle, rinse the wall of the intake tube with about 25 ml of 95 percent ethanol, collecting the rinse in the gas-washing bottle cylinder placed a few cm below its normal position.
b. Seat the cylinder in its normal position, and turn on the vacuum briefly, drawing any residual rinse through the bubbler and into the cylinder.
c. Remove the cylinder, and discard the rinse.
d. Repeat steps 4a. through 4c. with about 25 ml of purified water.
e. Repeat steps 4a. through 4c. with about 25 ml of 95 percent ethanol.
f. Seat the cylinder in its normal position, and turn on the vacuum to "dry" the inlet tube, bubbler tube and bubbler.
g. Replace the cylinder with a clean, dry one, and return the intake tube to its normal horizontal position. The ARS is now ready for the next sample.

5. Treat the "samples" and "standard" (duplicate) and 0.05N sulfuric acid "blank" identically as follows:
a. Pipet 5.00 ml into a 50 ml Centrifuge tube.
b. Add 3.00 ml of mercuric sulfate reagent, and mix.
c. Place in a water bath at 37°C for 10 minutes.
d. Allow to stand at room temperature for 30 minutes.
e. Add 250 ul of sodium nitrite solution, and mix by swirling occasionally for 30 minutes.

These are the "treated" "samples", "standard" and "blank".

6. Measure and record the absorbances at 495 nm of the "treated samples" and "treated standard" (duplicate) on a suitable spectrophotometer using matched, 1.00 cm cells with the "treated blank" as reference solvent.

7. Calculate the phenylephrine bitartrate delivered for each "sample" from the following formula, record and report.

Phenylephrine bitartrate (mg/spray) =

$$\frac{(Asam)\ (Wstd)\ (20)\ (1.56)}{(Astd)\ (500)\ (12)}$$

Asam, Astd = absorbances at 495 nm of the "treated sample" and the "treated standard" (average of duplicates), respectively.
Wstd = weight of phenylephrine hydrochloride taken for the "standard" (mg).
20 = total volume of the "sample" (ml).
500 = equivalent total volume of the "standard" (ml).
12 = number of sprays taken for the "sample" (spray).
1.56 = gravimetric factor, phenylephrine hydrochloride to phenylephrine bitartrate.

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Measure the absorbance of standard and sample solutions at 495 nm with the blank solution in the reference cell within 20 minutes after color development using a suitable spectrophotometer with matched 5.00 cm cells.

Calculate the amount of Phenylephrine Bitartrate delivered per spray.

4.7.3 Minimum number of doses. Prepare the immediate container, with the adapter, as directed in the labeling. Atomize the inhalation as directed in the labeling and determine the weight of every twentieth dose up to and including the 440th dose. Average the 22 dose weights. Determine the individual weights of the 441st through 450th doses. The average of the last 10 dose weights shall be not less than 97 percent of the average of the 22 dose weights and none of the last 10 dose weights shall be less than 90 percent of the average of the 22 dose weights.

4.7.4 Moisture. The total moisture content of the inhalation shall be determined as described in the J.A.Ph.A (Scientific Edition), Vol. 49, No. 2, in the article titled "Pressurized Pharmaceutical Aerosols for Inhalation Therapy II".

4.7.5 Tests for Phenylephrine Bitartrate.

4.7.5.1 Identification tests.

A. Dissolve about 3 gram of phenylephrine bitartrate in 15 ml of purified water, add 10 ml of 1.5N ammonium hydroxide and scratch the walls of the vessel until crystallization begins. Allow the mixture to stand in the refrigerator (10°C) for about 2 hours. Filter the mixture through a fritted glass filter of fine porosity and wash the precipitate with a few drops of chilled water and three 2 ml portions of chilled absolute ethanol. Dry the washed precipitate in vacuum over silica gel for 4 hours. The phenylephrine base so obtained meets the following specifications.

1. Melting range: 170°C to 177°C by the USP method, class 1.

2. Specific rotation: -51 to -57.5^{o} at 25°C/sodium D-line on a solution prepared by dissolving about 400 mg, accurately weighed, in sufficient 0.5N hydrochloric acid to make 10.00 ml.

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B. Dissolve about 5 mg of phenylephrine bitartrate in 1 ml of purified water and add 3 ml of 15 percent (w/v) mercuric sulfate in 5N sulfuric acid. Heat on a steam bath for 5 minutes, cool and add 3 ml of 0.1N sodium nitrite: a deep red color is produced.

C. Dissolve about 50 mg of phenylephrine bitartrate in 1 ml of purified water in a clean test tube and add 3 ml of 1.0N silver nitrate: a white precipitate is formed. Add sufficient 1.5N ammonium hydroxide to just dissolve the precipitate and heat in a steam bath for about 5 minutes: a silver mirror is formed on the wall of the test tube.

4.7.5.2 Assay.

Reagents:

Glacial acetic acid, reagent grade

Potassium acid phthalate, primary standard

Perchloric acid, standard 0.025N in glacial acetic acid - Dilute 2.1 ml of 70 percent perchloric acid and 5.0 ml of acetic anhydride to 1,000 ml with glacial acetic acid. Standardize by the assay procedure using about 20 mg of potassium acid phthalate.

Crystal violet indicator - Dissolve 20 mg of crystal violet in 100 ml of glacial acetic acid.

Procedure:

Dissolve about 30 mg, accurately weighed, of phenylephrine bitartrate in 2.0 ml of glacial acetic acid. Add 2 drops of crystal violet indicator and titrate with standard, 0.025N perchloric acid in glacial acetic acid to the disappearance of the violet color. Titrate a 2.0 ml glacial acetic acid blank in the same manner.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One package containing one vial equipped with metering valve and an oral adapter (plastic mouthpiece), as specified, constitutes one unit of issue.

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5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available vial equipped with a metering valve and an oral adapter (plastic mouthpiece) considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit packaging. Each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate packaging. Twelve (12) units of issue shall be packaged as specified in 5.2.14.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.3.6.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

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5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. In addition, Type I shelf life marking as specified in MIL-STD-129 shall be shown.

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall also be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-00-071-7861	ISOPROTERENOL HYDROCHLORIDE AND PHENYLEPHRINE BITARTRATE AEROSOL, 22.5 ml, 450 Metered Doses

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA - DM

Project No. 6505-1706

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE	
MIL-I-37559	ISOPROTERENOL HYDROCHLORIDE AND PHENYLEPHRINE BITARTRATE AEOOSOL, 22.5 ml, 450 Metered Doses

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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